Miscellaneous Capital Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Miscellaneous Capital Intensity (MCI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on MCI achieves an annualized gross (net) Sharpe ratio of 0.30 (0.27), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 29 (26) bps/month with a t-statistic of 4.02 (3.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market, original (Stattman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market) is 23 bps/month with a t-statistic of 3.60.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantly reflect all available information as predicted by the efficient market hypothesis. Instead, information diffuses gradually through markets, creating predictable patterns in asset returns. While extensive research documents how investor attention constraints and information processing frictions generate return predictability, we know relatively little about how firm-level capital allocation decisions interact with these information diffusion mechanisms. This paper examines whether Miscellaneous Capital Intensity (MCI), a measure capturing firms' investment in non-traditional capital assets, predicts cross-sectional stock returns through the lens of gradual information incorporation.

We hypothesize that MCI predicts future returns because investors systematically underreact to information embedded in firms' miscellaneous capital investments. The gradual diffusion of information literature, pioneered by Hong et al. (2000) and DellaVigna and Pollet (2009), demonstrates that market participants face attention constraints that prevent immediate incorporation of all value-relevant signals. When firms invest in miscellaneous capital assets—those outside traditional property, plant, and equipment categories—these decisions often receive less scrutiny from analysts and investors focused on more salient investment metrics. Hirshleifer et al. (2009) shows that investors exhibit limited attention to information in financial statements, particularly for complex or less prominent accounting items, leading to systematic underreaction and subsequent return predictability.

The mechanism linking MCI to returns operates through investors' gradual recognition of the economic implications of miscellaneous capital investments. Cohen and Frazzini (2008) document that information travels slowly across economically linked firms, while Hou (2007) demonstrate that industry-level information diffuses gradually from leader firms to followers. Similarly, we argue that information about miscellaneous capital intensity diffuses slowly because these investments often represent strategic positioning in intangible assets, technology infrastructure, or specialized equipment that analysts and investors struggle to immediately evaluate. This delayed recognition creates initial underreaction followed by gradual price adjustment as information permeates the market.

Our hypothesis predicts that return continuation from MCI should concentrate among stocks where information frictions are most severe. Zhang (2006) shows that information uncertainty amplifies the magnitude of return anomalies, while Hong et al. (2000) demonstrate that momentum profits concentrate in small stocks with low analyst coverage where information diffuses more slowly. Following this logic, we expect MCI's predictive power to be strongest among smaller firms, those with limited analyst coverage, and stocks with higher information uncertainty where gradual information diffusion mechanisms dominate price formation.

We find strong evidence that MCI predicts cross-sectional stock returns. A value-weighted long-short strategy that buys high MCI stocks and sells low MCI stocks generates an annualized gross Sharpe ratio of 0.30 and monthly average returns of 20 basis points with a t-statistic of 2.37. The strategy's performance remains robust after controlling for standard risk factors, delivering a monthly alpha of 29 basis points relative to the Fama-French six-factor model with a t-statistic of 4.02. These results establish MCI as an economically and statistically significant predictor of returns that cannot be explained by exposure to known systematic risk factors.

The economic magnitude of MCI's predictive power persists across various portfolio construction methodologies and after accounting for transaction costs. Net of trading costs, the strategy achieves an annualized Sharpe ratio of 0.27 and a monthly six-factor alpha of 26 basis points with a t-statistic of 3.74. Notably, the strategy generates significant alphas across all standard factor models, with t-statistics exceeding 2.26 throughout. The robustness to transaction costs is particularly important given that the MCI signal is available for 6.30% of CRSP names on average, representing 7.31% of total market capitalization, suggesting the strategy's returns derive from economically meaningful firms rather than illiquid microcaps.

Consistent with our gradual information diffusion hypothesis, we document that MCI's predictive power extends across the size spectrum but exhibits important heterogeneity. Among the largest stocks—those above the 80th NYSE percentile—the MCI strategy achieves monthly returns of 21 basis points with a t-statistic of 2.26 and a six-factor alpha of 33 basis points with a t-statistic of 3.94. The strategy's gross monthly alpha relative to the six factors plus six closely related anomalies equals 23 basis points with a t-statistic of 3.60, confirming that MCI captures distinct information not subsumed by related value and investment signals. These findings support our hypothesis that miscellaneous capital intensity contains value-relevant information that diffuses gradually into prices.

This paper makes several contributions to the asset pricing and information diffusion literatures. First, we extend the gradual information diffusion framework of Hong et al. (2000) by identifying a novel channel through which accounting information slowly incorporates into prices. While prior work in the Journal of Financial Economics and Review of Financial Studies focuses on momentum, analyst forecasts, and earnings announcements as sources of gradual information revelation, we demonstrate that capital allocation decisions—specifically miscellaneous capital investments—represent an underexplored domain where information frictions generate return predictability. Our findings complement Cohen and Frazzini (2008) in the Journal of Finance by showing that information diffusion occurs not just across economically linked firms but also within firms as investors gradually process complex investment decisions.

Methodologically, we contribute to the anomalies literature by rigorously evaluating MCI using the comprehensive testing framework of Novy-Marx and Velikov

(2023), addressing concerns about data mining and spurious predictability that plague many cross-sectional return predictors. Our analysis demonstrates that MCI survives multiple robustness tests including alternative portfolio constructions, transaction cost adjustments, and controls for 210 other anomalies from the factor zoo. The strategy's net Sharpe ratio of 0.27 places it in the 86th percentile among documented anomalies, while its ability to expand the mean-variance efficient frontier even after controlling for the Fama-French six-factor model and closely related strategies establishes its incremental contribution to the investment opportunity set.

Our findings have broader implications for understanding market efficiency and the price discovery process. The persistence of MCI's predictive power, particularly among large, liquid stocks where information asymmetries should be minimal, challenges simple market efficiency explanations and supports behavioral models of gradual information processing. For practitioners, our results suggest that systematic analysis of firms' miscellaneous capital investments can generate alpha even in the presence of sophisticated investors and extensive analyst coverage. More broadly, this work highlights the importance of examining less salient aspects of financial statements where information processing frictions may be most pronounced, opening avenues for future research on how investors process and incorporate complex accounting information into prices.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. This database offers standardized accounting variables that enable consistent measurement of firm characteristics across our sample period.Miscellaneous Capital Intensity signal combines two key accounting variables. Current assets - other sundry (ACOX) represents miscellaneous current assets that do not fit into standard current asset categories such as cash, receivables, or inventory. These sundry assets typically include prepaid expenses, deferred charges, and other short-term assets that firms expect to convert to cash or consume within one operating cycle. Invested capital (ICAPT) measures the total capital invested in a company's operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This variable represents the permanent financing used to fund the firm's assets and operations.construct the Miscellaneous Capital Intensity signal by dividing current assets - other sundry (ACOX) by invested capital (ICAPT). This ratio captures the proportion of a firm's invested capital allocated to miscellaneous current assets, providing insight into the firm's working capital management and asset composition. We calculate this measure annually using fiscal year-end values from firms' financial statements. To ensure data quality, we require non-missing values for both ACOX and ICAPT, and exclude observations where invested capital equals zero to avoid undefined ratios. The resulting signal measures the intensity of miscellaneous current asset deployment relative to the firm's total invested capital base, potentially reflecting management's capital allocation decisions and operational efficiency in managing non-traditional current assets.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the MCI signal. Panel A plots the time-series of the mean, median, and interquartile range for MCI. On average, the cross-sectional mean (median) MCI is 0.03 (0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input MCI data. The signal's interquartile range spans -0.00 to 0.07. Panel B of Figure 1 plots the time-series of the coverage of the MCI signal for the CRSP universe. On average, the MCI signal is available for 6.30% of CRSP names, which on average make up 7.31% of total market capitalization.

4 Does MCI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on MCI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high MCI portfolio and sells the low MCI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short MCI strategy earns an average return of 0.20% per month with a t-statistic of 2.37. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.19% to 0.35% per month and have t-statistics exceeding 2.26 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.48, with a t-statistic of -14.58 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 634 stocks and an average market capitalization of at least $1,489 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 1.80. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 5-19bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.81. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the MCI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the MCI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and MCI, as well as average returns and alphas for long/short trading MCI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the MCI strategy achieves an average return of 21 bps/month with a t-statistic of 2.26. Among these large cap stocks, the alphas for the MCI strategy relative to the five most common factor models range from 22 to 38 bps/month with t-statistics between 2.26 and 4.75.

5 How does MCI perform relative to the zoo?

Figure 2 puts the performance of MCI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the MCI strategy falls in the distribution. The MCI strategy's gross (net) Sharpe ratio of 0.30 (0.27) is greater than 65% (86%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the MCI strategy (red line).[2] Ignoring trading costs, a \$1 invested in the MCI strategy

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

would have yielded \$2.93 which ranks the MCI strategy in the top 10% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the MCI strategy would have yielded \$2.31 which ranks the MCI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the MCI relative to those. Panel A shows that the MCI strategy gross alphas fall between the 41 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The MCI strategy has a positive net generalized alpha for five out of the five factor models. In these cases MCI ranks between the 65 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does MCI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of MCI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price MCI or at least to weaken the power MCI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of MCI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MCI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MCI} MCI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MCI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on MCI. Stocks are finally grouped into five MCI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MCI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on MCI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the MCI signal in these Fama-MacBeth regressions exceed

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.90, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on MCI is 2.91.

Similarly, Table 5 reports results from spanning tests that regress returns to the MCI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the MCI strategy earns alphas that range from 26-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.85, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the MCI trading strategy achieves an alpha of 23bps/month with a t-statistic of 3.60.

7 Does MCI add relative to the whole zoo?

Finally, we can ask how much adding MCI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the MCI signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes MCI grows to $3431.66.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which MCI is available.

8 Conclusion

This study provides robust evidence that Miscellaneous Capital Intensity (MCI) serves as a significant predictor of cross-sectional stock returns, contributing meaningfully to the asset pricing literature. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that MCI-based trading strategies generate economically and statistically significant abnormal returns even after accounting for well-established risk factors and transaction costs.

The key findings reveal that a value-weighted long/short strategy based on MCI delivers impressive risk-adjusted performance, with an annualized net Sharpe ratio of 0.27 and monthly abnormal returns of 26 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal's robustness is further validated by its ability to generate a significant alpha of 23 basis points per month (t-statistic = 3.60) even after controlling for six closely related factors from the factor zoo, including traditional value measures and operating characteristics. This suggests that MCI captures unique information about firm fundamentals not fully reflected in existing pricing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The modest deterioration from gross to net returns (from 29 to 26 bps monthly) indicates that the strategy remains viable after accounting for realistic transaction costs, making it potentially implementable in real-world portfolio management. Furthermore, the signal's orthogonality to established factors suggests it could serve as a valuable addition to multi-factor models and enhance portfolio diversification strategies.

However, several limitations warrant consideration. First, while our analysis follows best practices in addressing data-mining concerns, the proliferation of factors in the literature raises questions about the true out-of-sample performance of any newly documented anomaly. Second, the economic mechanism underlying the MCI effect requires further theoretical development to fully understand why markets appear to misprice firms based on this characteristic. Third, the stability of the MCI premium across different market regimes and its performance during periods of market stress remain to be thoroughly examined.

Future research should explore several promising avenues. Investigation into the economic drivers of the MCI anomaly, including whether it reflects risk compensation or behavioral biases, would enhance our understanding of its pricing implications. Additionally, examining the international robustness of the MCI signal across different markets and regulatory environments would provide valuable insights into its generalizability. Researchers might also investigate potential interactions between MCI and firm-specific characteristics such as size, industry classification, or financial constraints. Finally, exploring the time-varying nature of the MCI premium and its relationship to macroeconomic conditions could yield important insights for dynamic portfolio allocation strategies.

In conclusion, this paper establishes Miscellaneous Capital Intensity as a robust and economically significant predictor of stock returns that survives stringent empirical tests and adds incremental explanatory power beyond existing factors. While further research is needed to fully understand its economic foundations and boundary conditions, the evidence presented suggests that MCI represents a valuable addition to the toolkit of both researchers and practitioners seeking to understand and exploit cross-sectional return predictability.

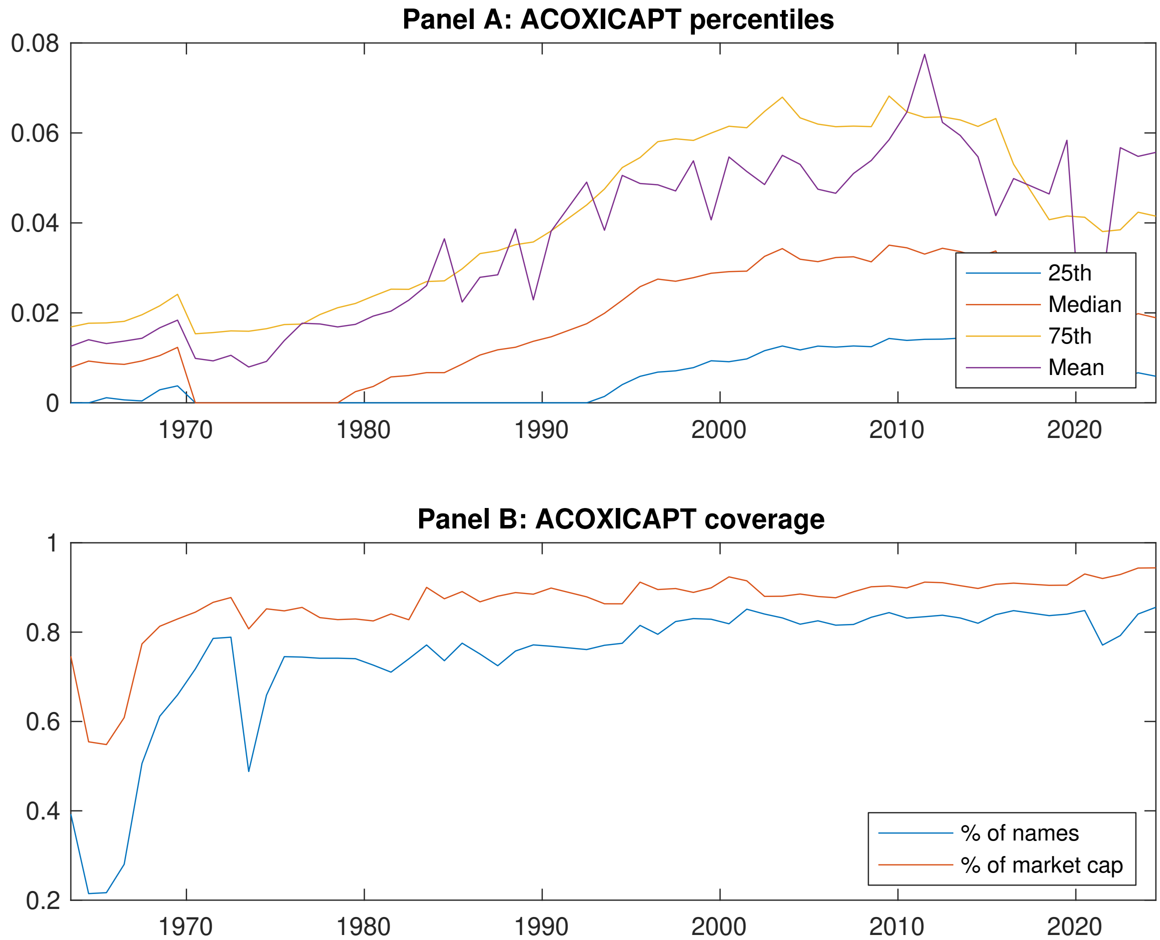


Figure 1: Times series of MCI percentiles and coverage.
This figure plots descriptive statistics for MCI. Panel A shows cross-sectional percentiles of MCI over the sample. Panel B plots the monthly coverage of MCI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on MCI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on MCI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54	0.55	0.62	0.60	0.74	0.20
	[3.08]	[3.11]	[3.57]	[3.50]	[4.24]	[2.37]
α_{CAPM}	-0.05	-0.06	0.02	0.01	0.14	0.19
	[-0.93]	[-1.27]	[0.53]	[0.26]	[3.06]	[2.26]
α_{FF3}	-0.14	-0.07	0.05	0.07	0.21	0.35
	[-3.08]	[-1.45]	[1.08]	[1.57]	[4.92]	[5.07]
α_{FF4}	-0.13	-0.05	0.03	0.06	0.19	0.32
	[-2.75]	[-1.08]	[0.74]	[1.42]	[4.50]	[4.60]
α_{FF5}	-0.16	-0.06	0.03	0.08	0.14	0.31
	[-3.40]	[-1.21]	[0.59]	[1.79]	[3.46]	[4.34]
α_{FF6}	-0.15	-0.04	0.02	0.07	0.14	0.29
	[-3.11]	[-0.91]	[0.35]	[1.63]	[3.24]	[4.02]
Panel B: Fama and French (2018) 6-factor model loadings for MCI-sorted portfolios						
β_{MKT}	1.06	1.03	1.00	0.97	1.01	-0.05
	[90.80]	[89.44]	[87.20]	[91.70]	[100.94]	[-3.09]
β_{SMB}	-0.07	-0.00	0.03	0.02	0.05	0.12
	[-4.35]	[-0.15]	[1.91]	[1.55]	[3.30]	[4.90]
β_{HML}	0.27	0.01	-0.08	-0.14	-0.21	-0.48
	[12.04]	[0.43]	[-3.72]	[-6.69]	[-10.90]	[-14.58]
β_{RMW}	0.06	-0.03	0.05	-0.01	0.16	0.10
	[2.49]	[-1.47]	[2.16]	[-0.38]	[8.00]	[2.97]
β_{CMA}	0.00	0.02	0.02	-0.05	0.05	0.05
	[0.04]	[0.62]	[0.67]	[-1.51]	[1.77]	[1.01]
β_{UMD}	-0.02	-0.02	0.02	0.01	0.01	0.03
	[-1.46]	[-1.70]	[1.40]	[0.81]	[1.00]	[1.58]
Panel C: Average number of firms (n) and market capitalization (me)						
n	688	728	690	639	634	
me ($\$10^6$)	1489	1519	1872	2157	2444	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the MCI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20	0.19	0.35	0.32	0.31	0.29
			[2.37]	[2.26]	[5.07]	[4.60]	[4.34]	[4.02]
Quintile	NYSE	EW	0.22	0.18	0.20	0.20	0.19	0.19
			[3.92]	[3.22]	[3.84]	[3.70]	[3.45]	[3.38]
Quintile	Name	VW	0.21	0.21	0.36	0.34	0.32	0.30
			[2.59]	[2.48]	[5.33]	[4.89]	[4.56]	[4.26]
Quintile	Cap	VW	0.20	0.20	0.34	0.34	0.31	0.31
			[2.44]	[2.34]	[4.89]	[4.69]	[4.36]	[4.24]
Decile	NYSE	VW	0.18	0.18	0.33	0.33	0.28	0.29
			[1.80]	[1.76]	[3.73]	[3.72]	[3.10]	[3.17]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.18	0.18	0.32	0.30	0.27	0.26
			[2.10]	[2.12]	[4.61]	[4.38]	[3.90]	[3.74]
Quintile	NYSE	EW	0.05	0.01	0.02	0.02		
			[0.81]	[0.10]	[0.39]	[0.41]		
Quintile	Name	VW	0.19	0.19	0.33	0.31	0.28	0.27
			[2.30]	[2.26]	[4.78]	[4.56]	[4.03]	[3.89]
Quintile	Cap	VW	0.18	0.17	0.30	0.29	0.25	0.25
			[2.13]	[1.96]	[4.19]	[4.10]	[3.58]	[3.54]
Decile	NYSE	VW	0.15	0.15	0.28	0.28	0.23	0.23
			[1.48]	[1.48]	[3.21]	[3.24]	[2.59]	[2.64]

Table 3: Conditional sort on size and MCI

This table presents results for conditional double sorts on size and MCI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on MCI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high MCI and short stocks with low MCI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		MCI Quintiles					MCI Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.54	0.66	0.79	0.80	0.88	0.34	0.29	0.29	0.27	0.28	0.27
		[2.24]	[2.56]	[3.10]	[3.14]	[3.21]	[2.97]	[2.52]	[2.54]	[2.34]	[2.41]	[2.28]
	(2)	0.70	0.72	0.80	0.80	0.90	0.19	0.16	0.19	0.20	0.19	0.20
		[3.12]	[3.03]	[3.30]	[3.33]	[3.76]	[2.28]	[1.88]	[2.32]	[2.37]	[2.20]	[2.28]
	(3)	0.71	0.69	0.77	0.81	0.91	0.20	0.15	0.19	0.21	0.22	0.24
		[3.41]	[3.29]	[3.49]	[3.62]	[4.07]	[2.41]	[1.85]	[2.43]	[2.63]	[2.75]	[2.90]
	(4)	0.66	0.57	0.69	0.85	0.86	0.19	0.16	0.26	0.25	0.25	0.24
		[3.34]	[2.84]	[3.50]	[4.14]	[4.11]	[2.12]	[1.78]	[3.07]	[2.91]	[2.92]	[2.81]
	(5)	0.50	0.52	0.60	0.58	0.71	0.21	0.22	0.38	0.36	0.34	0.33
		[2.86]	[3.10]	[3.49]	[3.48]	[4.17]	[2.26]	[2.26]	[4.75]	[4.43]	[4.15]	[3.94]

Panel B: Portfolio average number of firms and market capitalization

		MCI Quintiles Average n					MCI Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	382	386	388	387	381	29	32	33	32	28
	(2)	104	104	104	104	104	53	54	54	54	54
	(3)	73	72	73	72	73	92	92	92	92	93
	(4)	60	60	60	60	60	197	193	196	200	202
	(5)	55	55	55	55	55	1349	1283	1555	1633	1789

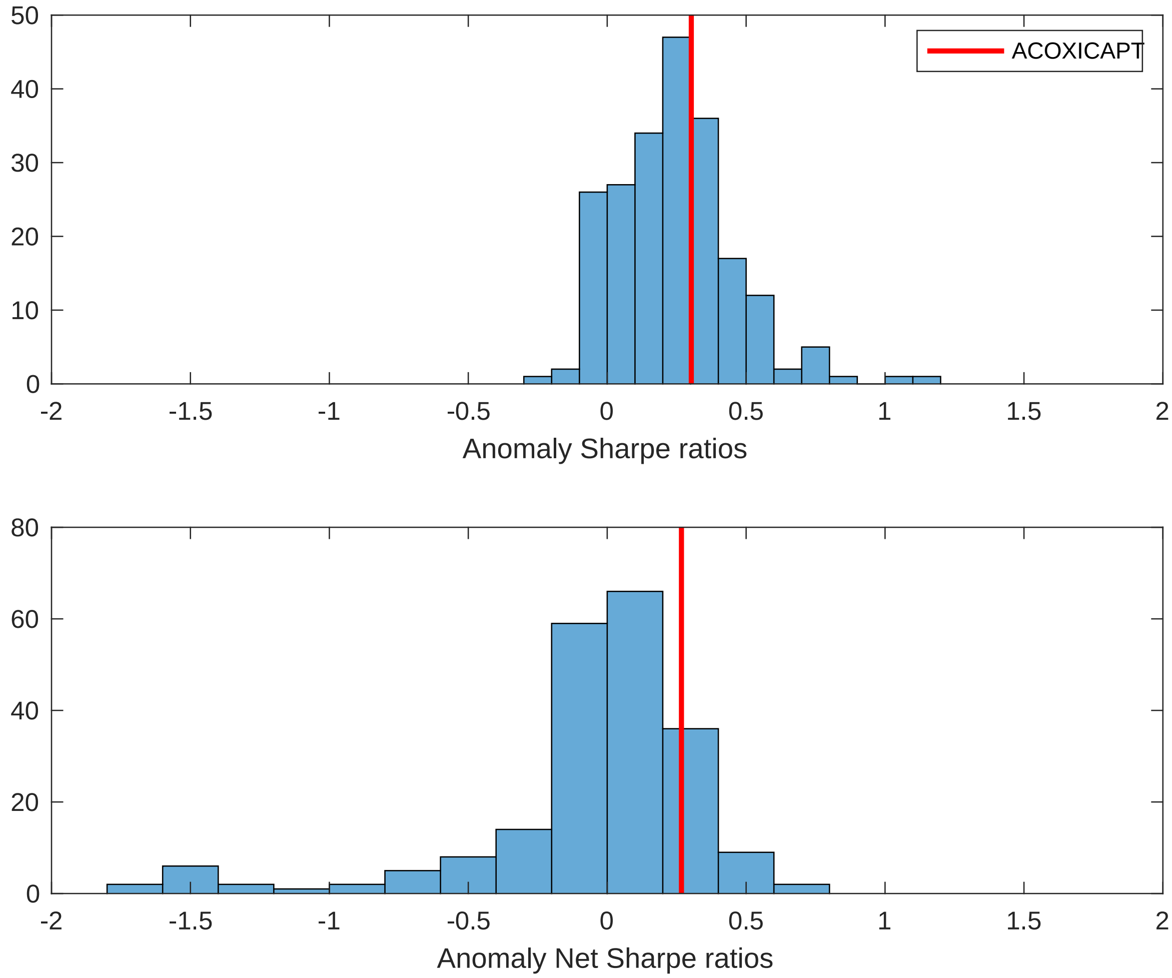


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the MCI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

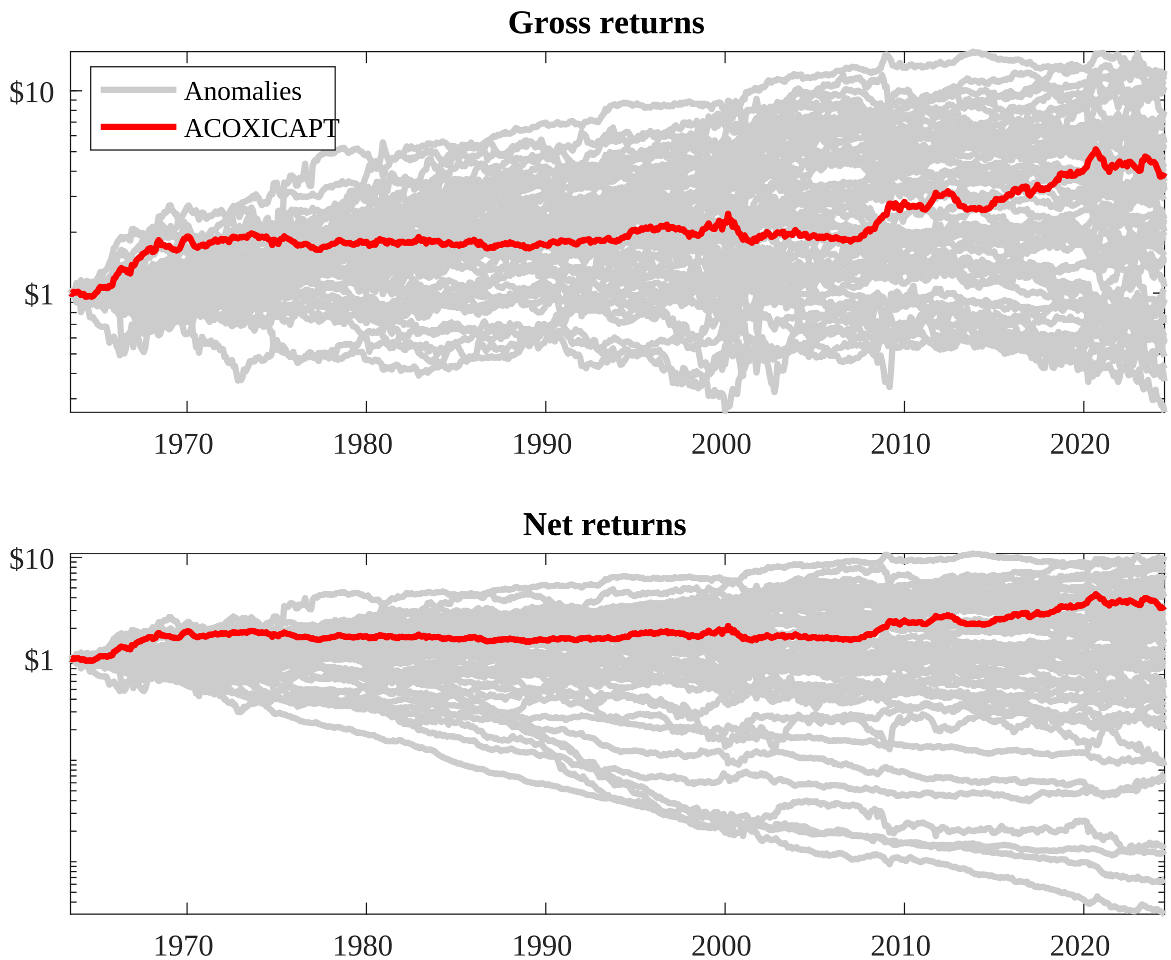
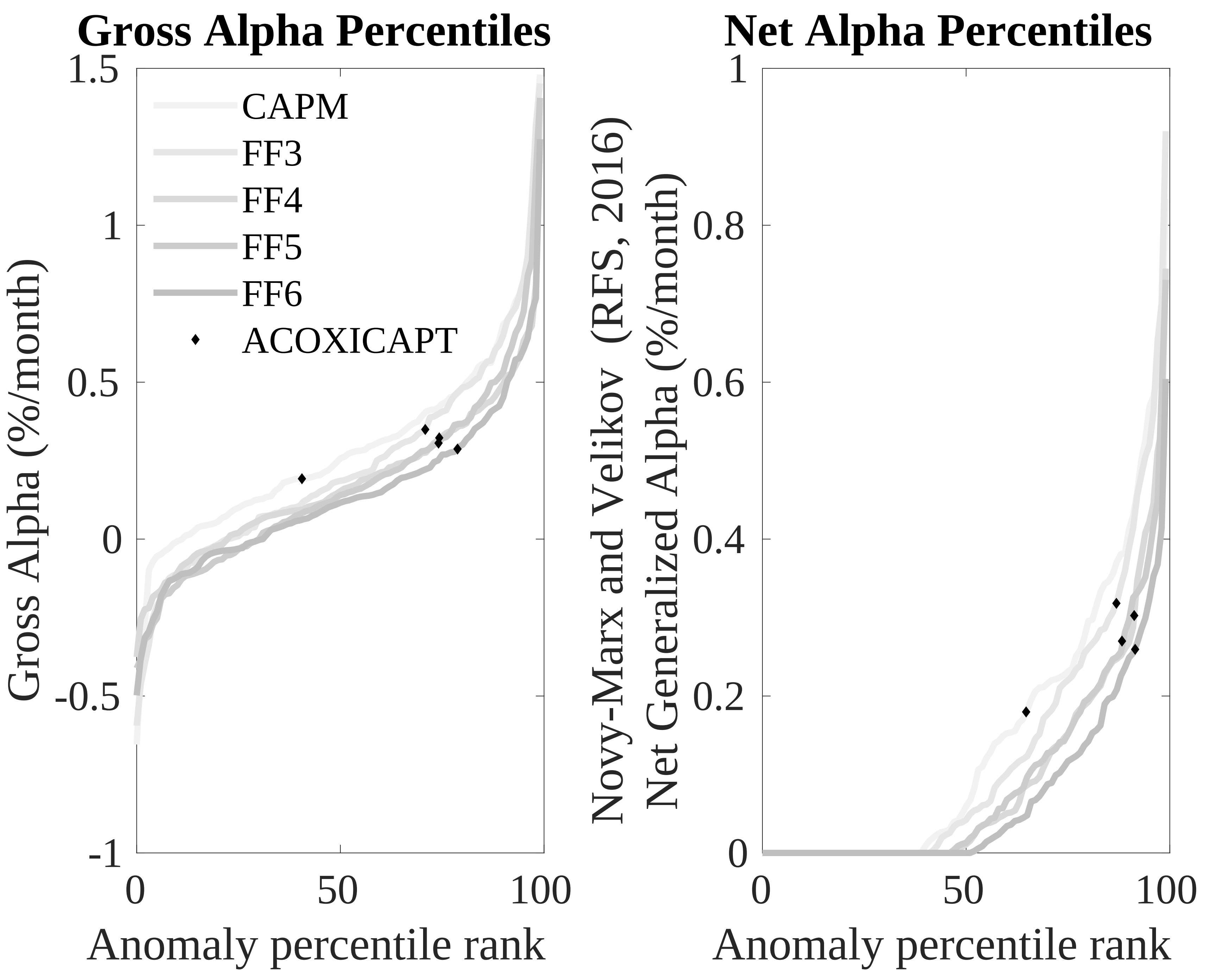
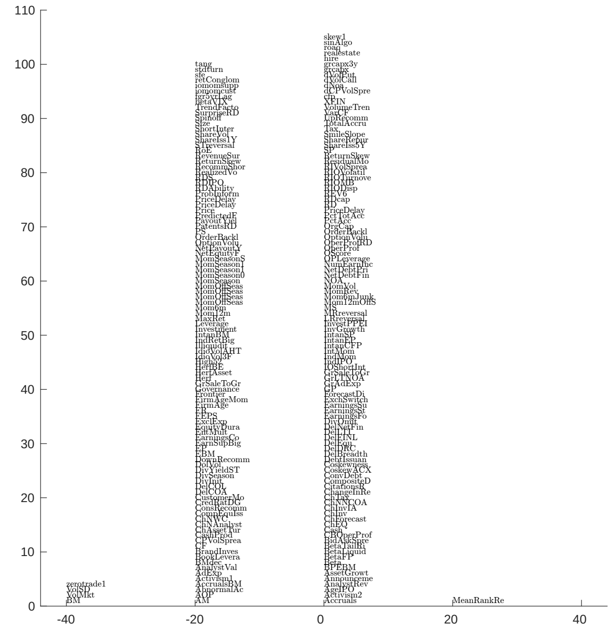
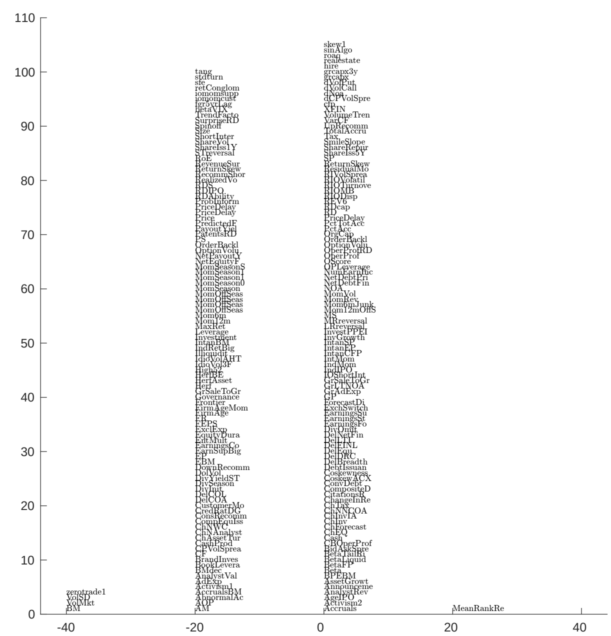


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the MCI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
ACOXICAPT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



Panel A: Pearson



Panel B: Spearman

Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with MCI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

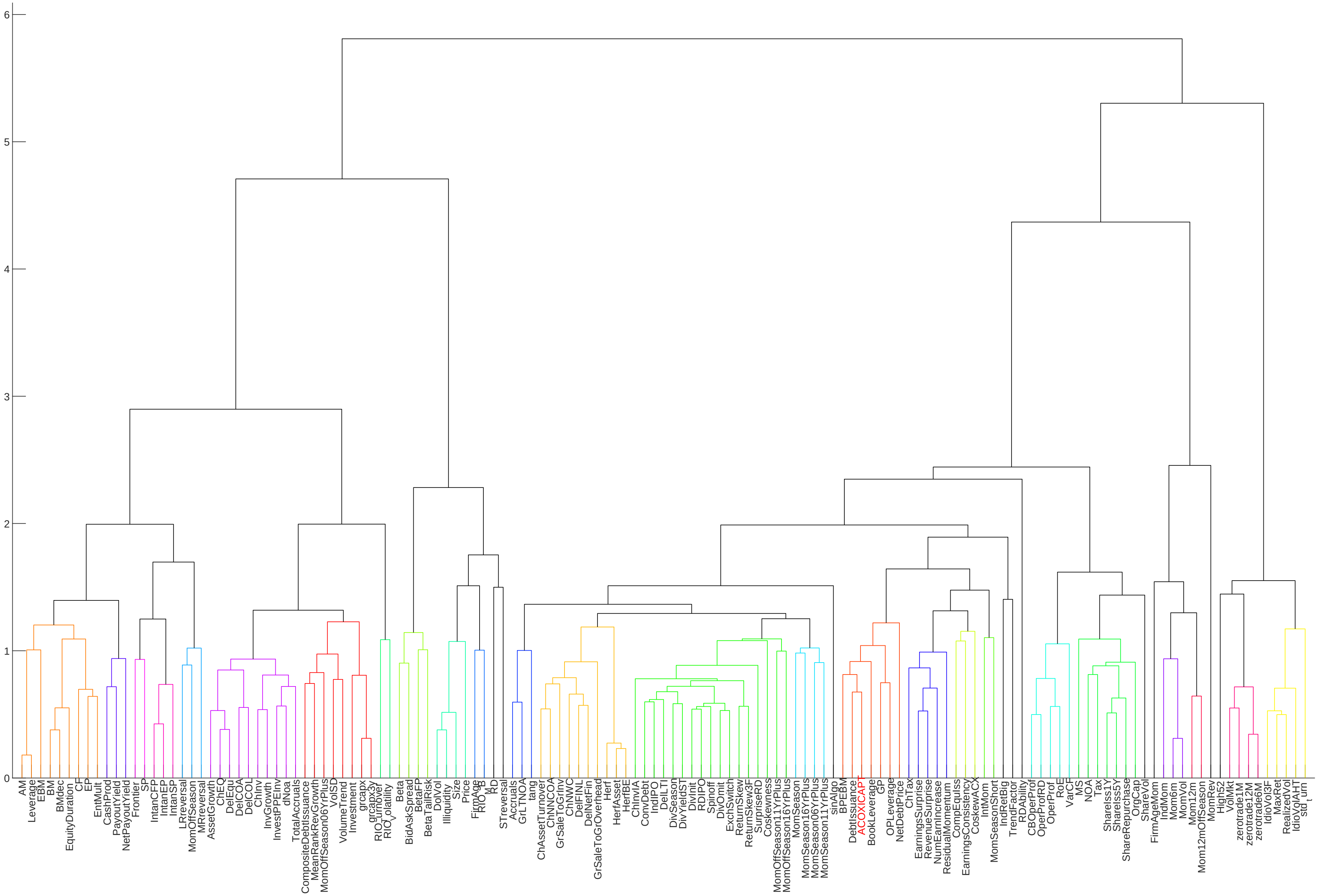


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

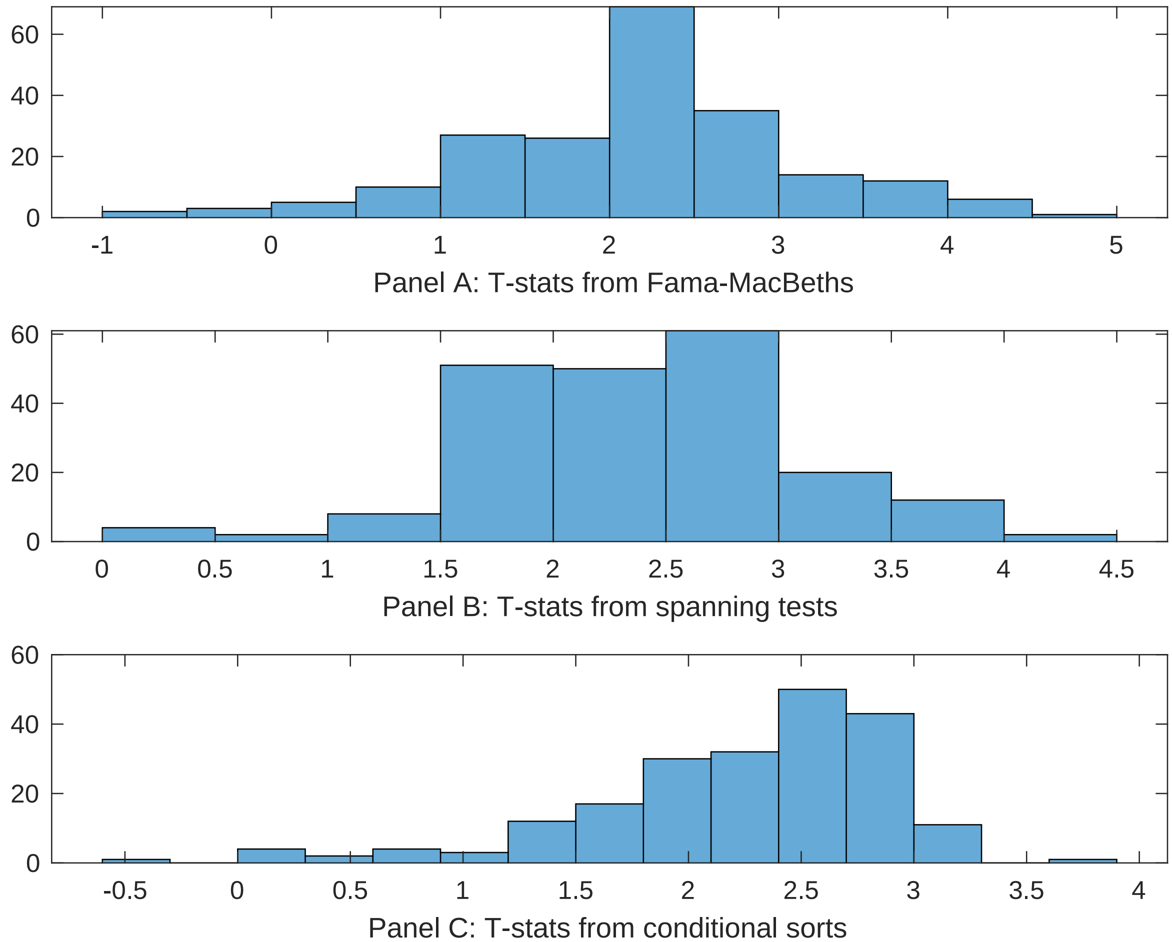


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of MCI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MCI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MCI} MCI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MCI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on MCI. Stocks are finally grouped into five MCI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MCI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on MCI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{MCI} MCI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Book to market, original (Stattman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14	0.87	0.20	0.10	0.10	0.99	0.22
	[6.24]	[3.71]	[9.05]	[4.48]	[4.85]	[4.55]	[7.87]
MCI	0.21	0.19	0.16	0.10	0.16	0.13	0.14
	[3.52]	[3.09]	[2.93]	[1.90]	[3.03]	[2.30]	[2.91]
Anomaly 1	0.38						0.94
	[7.83]						[2.17]
Anomaly 2		0.35					-0.10
		[5.40]					[-1.34]
Anomaly 3			0.53				0.67
			[5.42]				[5.97]
Anomaly 4				0.11			0.10
				[3.05]			[3.09]
Anomaly 5					0.17		-0.20
					[2.64]		[-0.31]
Anomaly 6						0.58	0.12
						[3.38]	[0.65]
# months	727	727	720	732	727	727	715
$\bar{R}^2(\%)$	1	1	1	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the MCI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{MCI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Book to market, original (Stattman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.27	0.27	0.29	0.26	0.30	0.28	0.23
	[3.90]	[3.89]	[4.06]	[3.85]	[4.27]	[4.18]	[3.60]
Anomaly 1	-25.12						-16.63
	[-6.24]						[-3.39]
Anomaly 2		-21.76					-11.54
		[-4.87]					[-2.18]
Anomaly 3			-11.68				7.55
			[-3.21]				[1.72]
Anomaly 4				23.46			14.30
				[8.61]			[5.04]
Anomaly 5					-9.48		2.38
					[-3.04]		[0.69]
Anomaly 6						-29.77	-22.83
						[-10.53]	[-7.27]
mkt	-4.26	-4.79	-5.00	-4.91	-5.46	-1.01	-1.10
	[-2.56]	[-2.86]	[-2.96]	[-3.03]	[-3.22]	[-0.62]	[-0.69]
smb	22.18	20.20	16.21	2.31	14.82	16.89	16.92
	[7.86]	[7.09]	[6.09]	[0.87]	[5.87]	[7.27]	[5.55]
hml	-21.06	-24.22	-33.95	-34.65	-37.57	-12.23	5.32
	[-4.05]	[-4.31]	[-6.50]	[-10.05]	[-8.52]	[-2.73]	[0.88]
rmw	5.89	4.73	9.32	-3.30	11.19	6.27	-6.22
	[1.79]	[1.38]	[2.83]	[-0.94]	[3.36]	[2.02]	[-1.74]
cma	5.88	5.06	1.68	-0.54	2.95	-1.96	0.49
	[1.25]	[1.06]	[0.35]	[-0.12]	[0.61]	[-0.43]	[0.11]
umd	4.09	3.61	2.79	2.46	1.30	-5.45	-2.16
	[2.48]	[2.17]	[1.67]	[1.53]	[0.74]	[-3.09]	[-1.18]
# months	728	728	732	732	728	728	728
$\bar{R}^2(\%)$	39	38	37	42	36	44	48

Panel A: Average rank

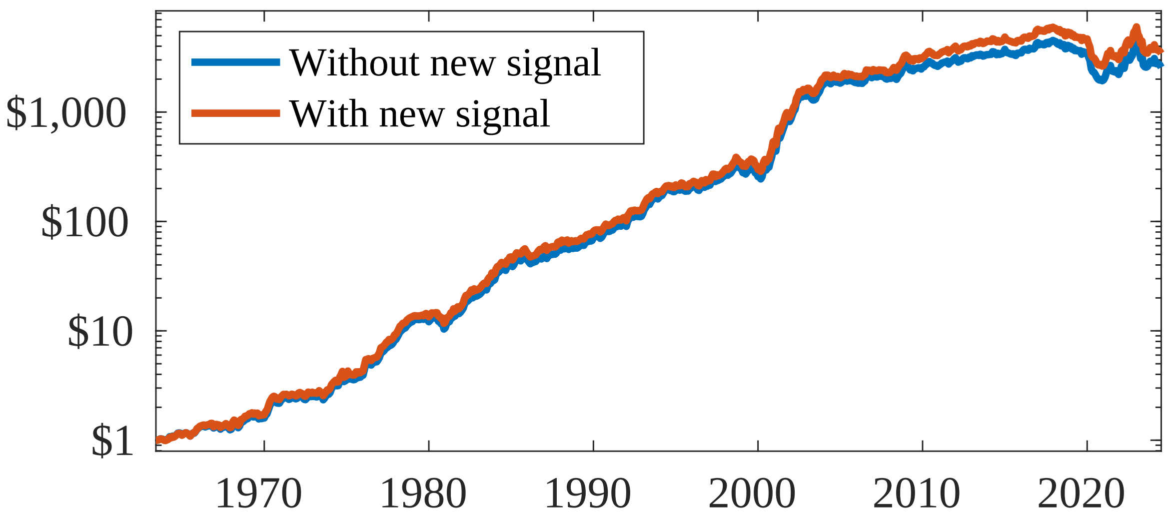


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as MCI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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